



Indian Association for the Cultivation of Science
(Deemed to be University under the *de novo* category)
Integrated Bachelor's-Master's Program
End-Semester Examination (Sem-I) - Autumn 2025

Subject: Mathematics I

Subject Code: MAT 1101

Full marks: 50

Time allotted: 3 hours

- Questions 1-2 together carry a total of 10 marks, while Questions 3-7 together carry a total of 50 marks.
- You may attempt as many as you want, however, in Questions 3-7, the MAXIMUM you can score is 40 marks.

1. For each of the following statements, simply say whether it is **True** or **False**.
No explanation/justification is necessary! (6 × 1) = [6]

- (a) If $f : A \rightarrow B$ is one-one, then $f(C \cap D) = f(C) \cap f(D)$ for any $C \subset A, D \subset A$.
- (b) Every bounded non-empty set of real numbers has a maximum and a minimum.
- (c) If $\sup A < \sup B$, then there must exist $b \in B$ which is an upper bound for A .
- (d) If a sequence of integers converges, then the limit must be an integer.
- (e) If a sequence of irrationals converges, then the limit must be an irrational.
- (f) If (x_n) diverges to $-\infty$, then $(\sqrt{|x_n|+1}/x_n)$ must converge to 0.

2. Give, in one sentence, the statement of each of the following: (2 × 2) = [4]

- (a) Completeness Axiom for real numbers.
- (b) Definition of convergence a real sequence to a limit.

3. Show that the inequality $3^n > 2n$ holds for all $n \in \mathbb{N}$. [8]

4. Identify the set of all real numbers x that satisfy $|x+1| > 7 - |x-2|$. [8]

5. (a) Given two real numbers $x < y$, show exactly how you can always get a natural number n and an integer m , such that, $x < \frac{m}{n} < y$.

(b) For each real number x , denote $A(x) = \{r \in \mathbb{Q} : r < x\}$. Using the above (or otherwise), show that

(i) if $x < y$ are any two real numbers, then $A(x) \subsetneq A(y)$, and,

(ii) $\sup A(x) = x$, for any real number x .

(5 + (2 + 3)) = [10]

6. Find the limits of the following sequences, **clearly showing the steps**:

(You may use algebra of limits and also all standard limits proved in class)

(a) $\left(\frac{2 \cdot 7^n + (-2)^{n+1} + 5n^3}{7^{n+1} + 2^{n+2}} \right)$

(b) $(\sqrt{4n^2 + 3n} - 2n)$

(c) $((n!)^{1/n^2})$ [May use Squeeze Theorem.] $(3 \times 6) = [18]$

7. Consider the sequence (x_n) defined recursively by $x_1 = 3$ and $x_{n+1} = \frac{1}{2} \left(x_n + \frac{6}{x_n} \right)$, for $n \geq 1$. Provide the necessary and appropriate arguments to prove that the sequence (x_n) converges to a limit and also find the limit. [6]